

3rd Grade Biomes

(Scheduled Event - 30 Minutes)



Description: Participants will be assessed on their understanding of ecological principles, ecosystems, species interactions, and major terrestrial and aquatic biomes.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Teams must bring:

- Pen/Pencil

Teams may not bring:

- Notes or guides of any kind.

(Optional) Teams may bring:

- Four-function calculator

Safety Requirements: None

The Competition: This event may be administered as a written test or as a series of stations which can include diagrams, maps, specimens, photographs, data analysis, and problems for students to solve.

Content topics will include but are not limited to:

1. Ecology Fundamentals

- Define and apply the concepts of organism, population, community, ecosystem, biome, habitat, and niche.
- Distinguish between biotic and abiotic factors.
- Explain the roles of producers, consumers, and decomposers and how energy moves through ecosystems using food chains, food webs, and energy pyramids.
- Describe common species interactions including predation, competition, mutualism, commensalism, and parasitism.

2. Biomes and Adaptations

- Identify major terrestrial biomes including tundra, taiga, temperate deciduous forest, temperate grassland, desert, tropical rainforest, and savanna, as well as major freshwater and marine ecosystems.
- Describe the climate, vegetation, characteristic organisms, and geographic distribution of each biome.
- Explain how structural, behavioral, and physiological adaptations help organisms survive in specific environments.
- Compare and contrast the characteristics of different biomes and ecosystems.

3. Biodiversity and Human Impact

- Explain the importance of biodiversity and ecosystem health.
- Describe the effects of habitat loss, pollution, invasive species, and climate change on ecosystems and biomes.

- Identify conservation practices that help protect ecosystems and biodiversity.

4. Data Analysis and Applications

- Interpret diagrams, maps, charts, graphs, tables, and models related to ecology and biomes.
- Apply ecological principles and biome characteristics to analyze data and solve problems.

Examples:

- Given a food web, identify producers, consumers, predators, prey, and decomposers.
- Interpret an energy pyramid to determine which trophic level contains the most energy.
- Compare two biomes using climate graphs, maps, or data tables.
- Use a biome map to identify the most likely biome for a given location.
- Analyze population graphs to identify trends, limiting factors, or carrying capacity.
- Interpret predator-prey graphs to determine how changes in one population affect another.
- Use ecological succession diagrams to identify the stage of ecosystem development.
- Interpret species distribution maps to predict the effects of habitat loss or fragmentation.
- Analyze tables of temperature, precipitation, and vegetation to identify a biome.
- Predict how changes to one component of an ecosystem (e.g., invasive species, drought, wildfire, pollution) affect the rest of the ecosystem.
- Apply concepts of biodiversity, food webs, and ecological interactions to solve ecosystem-based scenarios.

Scoring:

- Points will be awarded for each correct response.

Tiebreakers:

- Ties will be broken by the accuracy or quality of answers to select questions chosen by the event leader prior to the competition.

Resources:

- [Biomes](#)
- [Ecology](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles